

AP Statistics Handout Key: Lesson 9.1

Topics: problem of multiple tests, chi-square test for goodness of fit

Lesson 9.1 Guided Notes

2014: Harvard sued by a group of Asian-American applicants who were rejected in admissions. They claimed racial discrimination.

2018: Courts released reports on Harvard admissions data, giving unprecedented view into admissions. The data we'll look at today is reconstructed from parts of the plaintiff report¹ where the defense and plaintiffs generally agreed on the findings.

2023: Supreme Court considered the case and found evidence of discrimination. The decision overturned the use of race-conscious admissions at most Universities.



Today's Key Analysis: Based on the data from the case, was there convincing evidence that Harvard discriminated against Asian applicants?

The problem of multiple tests

Class of 2019 Applicants,
Top 10% Academic Index

Group	%
Asian-American	57.5%
Hispanic	3.1%
African-American	0.8%
White	38.7%

Source: Plaintiff's report, table B.5.7

Class of 2019
Admitted Students

Group	%
Asian-American	21.4%
Hispanic	12.2%
African-American	11.2%
White	55.3%

Source: *Harvard Gazette*

Notes:

- Academic Index ratings are internal measures of academic qualification produced by the Harvard admissions office. They're calculated based on standardized test scores and high school grades/performance.
- Other groups (Native American, Mixed Race, etc.) and international students were not included in the court's main analysis. Percentages are calculated just out of these 4 groups.

Imagine Harvard claims: "We only accept the top academic applicants and we treat those applicants equally. Our admitted class is as good as a random sample from the pool of top applicants." You'd like to test if there's convincing evidence against this claim. Would it be appropriate to do four z-test for proportions (one for each racial group above)? Why or why not?

No! First, it'd be inconvenient to do so many tests. But more importantly: performing many tests compounds the probability of making a Type I error. The probability of falsely rejecting H_0 for any test is $\alpha = 0.05$ (usually). So, you make a mistake 5% of the time, or 1 in 20 times. If you do many tests, you will make an error eventually!

¹ Plaintiff report accessible here: https://drive.google.com/file/d/1eh4aOz1O7iO0ICQB99XYMIWoCdpoSCC/view?usp=share_link

Chi-Square test for goodness of fit

Hypotheses

H_0 : The racial distribution of admitted students is the same as the claimed distribution

H_A : The racial distribution of admitted students is not the same as the claimed distribution

The null (dull/default/innocent) hypothesis: there is no racial discrimination, so the distribution of admitted students matches the population of top applicants.

The alternative (guilty/proof-needed) hypothesis: the distribution of admitted students does not match the population of top applicants.

Logic of Hypothesis Tests

1. Start by assuming the null (the default) is true. *In your own words, describe what that means in context of the example above:*

Assume Harvard's admissions process is similar to randomly sampling from the distribution of top academic applicants

2. Ask yourself: if the null is *in fact true*, how likely is the data that you've gathered? *In your own words, describe what that means in context of the example above:*

Compare the racial distribution in the sample of admits to the population of top academic applicants.

3. Draw a conclusion. *In your own words, describe what that means in context of the example above:*

If different enough, conclude that admitted students are not a random sample from the pool of top academic applicants.

Calculating the Test Statistic: χ^2

In total, Harvard admitted $n = 2023$ applicants from these racial groups to the Class of 2019. How many would we **expect** to admit if it was a random sample?

Population of Top Academic Applicants	Group	%	Expected Counts
	Asian-American	57.5%	$(0.575) * 2023 = 1163.2$
	Hispanic	3.1%	$(0.031) * 2023 = 62.7$
	African-American	0.8%	$(0.008) * 2023 = 16.2$
	White	38.7%	$(0.387) * 2023 = 782.9$

Step 1: Assuming the null is true, this is what we'd expect

Step 2: Compare our null expectation to the actual data

Group	Expected Count	Observed Count	$\frac{(Observed - Expected)^2}{Expected}$
Asian-American	1163.2	432	$\frac{(432 - 1163.2)^2}{1163.2} = 459.6$
Hispanic	62.7	247	$\frac{(247 - 62.7)^2}{62.7} = 541.7$
African-American	16.2	226	$\frac{(226 - 16.2)^2}{16.2} = 2717.0$
White	782.9	1118	$\frac{(1118 - 782.9)^2}{782.9} = 143.4$

a) Why do we subtract the observed and expected values?

To compare what we actually observed to what we expected under the null assumption. The greater the difference, the more likely we are to reject H_0

b) Why do we square the differences?

To get rid of negatives, so they don't cancel out other values when we add up all the differences.

c) Why do we divide by the expected values?

To scale the differences by what we expected.

Formula:

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

Calculated with our data:

$$\chi^2 = 3861.3$$

When chi-square is **big**, that means your observed data is very **different** from what was expected under H_0 . Therefore, if H_0 is true, your observations are unlikely (**low p-value**). You reject H_0 .

When chi-square is **small**, that means your observed data is very **similar** to what was expected under H_0 . Therefore, if H_0 is true, your observations are likely (**high p-value**). You fail to reject H_0 .

Step 3: Conclude → We'll go through that in the four step process below!

The Four Steps Process

Imagine that Harvard claims: "We only accept the top academic applicants and we treat those applicants equally. Our admitted class is as good as a random sample from the pool of top applicants." Do the data below provide **convincing evidence** to refute Harvard's admissions claim?

Top Academic Applicants	Group	%
	Asian-American	57.5%
	Hispanic	3.1%
	African-American	0.8%
	White	38.7%

Admitted	Observed
	432
	247
	226
	1118

Show your work on the next page!

State:

H_0 : The racial distribution of admitted students is the **same** as the claimed distribution

$\alpha = 0.05$

H_A : The racial distribution of admitted students is **not the same** as the claimed distribution

Plan:

We will conduct a **chi-square test for goodness of fit**, if conditions met.

Random: We are assuming that the admitted students were selected randomly from the top academic applicant pool. ✓

10%: $n \leq 0.10N$

$2023 \leq 0.10$ (all top applicants)

May not be <10% of all top applicants – we'll proceed with caution ⚠

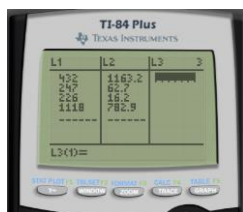
Large Counts:

All expected counts are ≥ 5 ✓

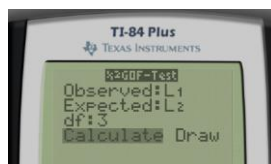
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Teacher note: On AP Exam FRQs, if unsure if condition met, write "proceed with caution" and conduct test. Don't want to lose points for not conducting the test!

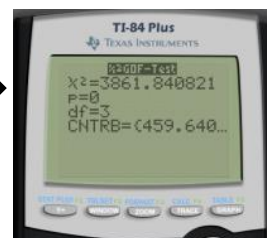
For "DO" phase: Calculator steps



STAT → EDIT →
L1: Observed
L2: Expected



STAT → TESTS →
D: X² GOF-TEST
Observed: L1
Expected: L2
Calculate



Output

Do:

Expected counts given above.
Degrees of freedom = $n_{group} - 1$
Degrees of freedom = $4 - 1$
Degrees of freedom = 3

$$\chi^2 = \frac{(432 - 1163.2)^2}{1163.2} + \frac{(247 - 62.7)^2}{62.7} + \dots$$

$$\chi^2 = 3861.8 \quad p\text{-value} \approx 0$$

Conclude:

Because our p-value (≈ 0) is **lower** than our α value (0.05), we **reject H_0** . We have convincing evidence that the racial distribution of admitted students is **different** than that of the top academic applicants.

Lesson 9.1 Discussion

Recommended discussion norms:
skewthescript.org/discussion-norms

The evidence presented so far:

1. Asians tended to be admitted at lower rates than expected, based on academic ratings.
2. Additionally: Asian applicants tended to receive² better extracurricular ratings from Harvard, on average, compared to the other racial groups.

Discussion Question: Imagine the plaintiffs argue that the above evidence proves racial discrimination. If you were Harvard's lawyer, how might you have attempted to defend the school's admissions policy? Explain.

Common Argument 1: There are more criteria than academics and extracurriculars

"If our admissions process only took into account academic and extracurricular achievement, there may be a case for discrimination. However, our admissions process is more holistic. We also take into account:

- Challenges and adversity an applicant has overcome.
- The strength of the high schools our applicants attend (a 1600 SAT score from a prestigious private school applicant may be less impressive than a 1550 from an applicant who attends a low-performing public school)
- Recommendation letters from teachers and counselors.
- Personality fit, as assessed by alumni interviews.
- Strength of personal statement and application essays.

Asian-American applicants may have the highest academic and extracurricular ratings on average, but when looking at individuals with the highest *overall* ratings on *all* of the above criteria, the racial distribution may look more similar to who was admitted."

Common Argument 2: The educational value of representing diverse perspectives

"We believe the education of our students is enhanced by attending a school in which a diversity of perspectives (as informed by a diversity of backgrounds) is represented. Asian-American students are already overrepresented in our school (about 5% of U.S. citizens are Asian, but about 20% of Harvard's Class of 2019 was Asian). There is educational value in promoting cross-cultural learning by making our class more broadly representative of the makeup of the country itself. That's why race is considered as one of many factors in the admissions process."

Important teacher notes:

- Two competing statistical papers – one completed by an expert for the plaintiffs and another by an expert for Harvard – aim to estimate (using logistic regression) if there is racial bias after controlling for academic ratings, extracurricular ratings, strength of recommendation letters, high school context, etc. The two authors disagree on the metrics used to estimate students' socioeconomic status and high school quality. They also disagree on whether the analyses should be done within years or across years. Thus, they arrive at difference conclusions: one finds evidence of bias after controlling for applicant factors, the other does not.
- In discussions of affirmative action, people tend to focus on whether Hispanic, Black, and Native American students are given racial preference. Two notes on this:

² See page 41 of plaintiff statistical report - https://drive.google.com/file/d/1eh4aOz1O7iO0lCQB99XYMIWoCdpoeSCC/view?usp=share_link



- Students from these racial backgrounds also disproportionately attend lower-performing high schools. So, what looks like racial preference may actually be preference given to the ability to perform well despite attending a low-performing high school.
- In the Harvard case, the plaintiffs' statistical expert [found](#) that athlete preference, legacy preference, and child-of-faculty preference disproportionately aided underqualified White applicants to Harvard. Many have called these preferences "affirmative action for White applicants."
- The Supreme Court ultimately [sided](#) with the plaintiffs, finding that Harvard discriminated against Asian-American applicants.

Most important note:

Talking about admissions can make many high school students feel they are undeserving of attending colleges. This can be especially true of Black, Hispanic, and Native American students. It's crucial to note that admissions is messy. For example, as mentioned above, preferences for legacy students, athletes, and children-of-faculty can sometimes function as "affirmative action for white applicants." Most people can probably find reasons to think their applications were given some sort of "unfair" preference.

While admissions statistics are worth studying, it's also important to note that statistical study of someone's "potential" will always be, to some extent, imperfect. So will admissions. Importantly, data is about trends, rather than individuals. Being admitted or not being admitted to a school is a far-from-perfect measure of an individual's merit or potential.

Lesson 9.1 Practice

1) Filip has 3D printed a 6-sided die. After rolling the die over 100 times, he wants to check that the distribution of outcomes for the die is uniform (i.e. the die is fair).

- a) Explain to Filip why a Chi-square test for goodness of fit would be appropriate here.

[Filip is observing the distribution of a categorical variable, and he wants to see if it significantly differs from a known uniform distribution.](#)

- b) Write the hypotheses for this procedure.

H_0 : The outcomes for the die are all equally likely.

H_A : The outcomes for the die are not all equally likely.

2) A large tech company offers a variety of work settings for its employees including in-person, remote, and hybrid (working in-person for part of the week and remote for the other days). The company claims that a large majority of its employees work remotely or in a hybrid setting, citing that 45% are remote and 30% are hybrid, while the rest work in-person. A random sample of 90 employees from this large company showed that 32 work remotely, 28 are hybrid, and 30 work in-person.

- a) If we want to investigate whether the sample data provides convincing evidence that the distribution of work settings for all employees is not the same as the company claims, what would our null and alternative hypotheses be?

H_0 : The work setting distribution for employees of this company is the same as the claimed distribution.

H_A : The work setting distribution for employees of this company is not the same as the claimed distribution.

- b) Assuming the null hypothesis is true, calculate the expected counts among the different work settings.

Remote: $90(0.45) = 40.5$

Hybrid: $90(0.3) = 27$

In-person: $90(0.25) = 22.5$

- c) How can we know if the observed distribution of work settings differs significantly from the expected distribution in part (b)?

Calculating chi-square will quantify the difference between the observed distribution and the expected distribution. The bigger chi-square is, the more these distributions differ. Then we use a chi-square distribution to calculate the probability of observing a chi-square that is that large or larger by chance alone (the p-value). If this probability is low enough, we say that we have convincing evidence that the distribution of work settings differs from the claim.

- d) How many degrees of freedom would be used for the chi-square distribution in this case?

$$df = n_{groups} - 1 = 3 - 1 = 2$$

- e) Assume the conditions for inference are met. Calculate the chi-square test statistic and the p-value, then write an appropriate conclusion using a 5% level of significance.

L1: 32, 28, 30

L2: 40.5, 27, 22.5

Chi-Square GOF Test(Observed: L1, Expected: L2, df: 2)

$$X^2 = 4.321 \quad p\text{-value} = 0.115$$

Because the p-value (0.115) is greater than our alpha level (0.05), we fail to reject H_0 . We do not have convincing evidence that the work setting distribution for all employees differs from the company's claim.

3) Multiple Choice: A multiple-choice exam has 100 questions with choices A, B, C, D, and E, and each choice is expected to be equally likely as the correct answer. The table below shows how often each choice is the correct answer.

Outcome	A	B	C	D	E
Frequency	16	24	26	19	15

Which of the following correctly calculates the chi-square test statistic from these outcomes?

- A) $X^2 = \frac{(16-20)^2}{20} + \frac{(24-20)^2}{20} + \frac{(26-20)^2}{20} + \frac{(19-20)^2}{20} + \frac{(15-20)^2}{20}$
- B) $X^2 = \frac{(16-20)^2}{16} + \frac{(24-20)^2}{24} + \frac{(26-20)^2}{26} + \frac{(19-20)^2}{19} + \frac{(15-20)^2}{15}$
- C) $X^2 = \left(\frac{16-20}{20} + \frac{24-20}{20} + \frac{26-20}{20} + \frac{19-20}{20} + \frac{15-20}{20} \right)^2$
- D) $X^2 = \frac{(16-20)^2}{100} + \frac{(24-20)^2}{100} + \frac{(26-20)^2}{100} + \frac{(19-20)^2}{100} + \frac{(15-20)^2}{100}$
- E) $X^2 = \frac{(16-0.2)^2}{100} + \frac{(24-0.2)^2}{100} + \frac{(26-0.2)^2}{100} + \frac{(19-0.2)^2}{100} + \frac{(15-0.2)^2}{100}$

4) A decade ago, the age distribution for users of a social media platform called My Face was as follows:

Age Range	Under 18	18-25	26-40	40-60	Over 60
Relative Frequency	0.15	0.73	0.06	0.03	0.03

The marketing executive of My Face is looking to see if the age distribution currently differs from this distribution. A random sample of 350 of the >1 million total users is selected. The age distribution for the sample is shown in the following table.

Age Range	Under 18	18-25	26-40	40-60	Over 60
Frequency	22	59	110	114	45

Do these results provide convincing evidence that the age distribution for all current users of My Face is different from the distribution a decade ago?

State: H_0 : The age distribution for all current users of My Face is the same as a decade ago.

H_A : The age distribution for all current users of My Face is not the same as a decade ago.

$\alpha = 0.05$

Plan: We will conduct a Chi-square test for goodness of fit, if conditions are met.

Random: The My Face user sample was randomly selected from all users. ✓

10%: $350 \leq 0.10(\text{over 1 million users}) \rightarrow$ This is clearly true ✓

Large Counts: All expected counts are at least 5. ✓

Age Range	Under 18	18-25	26-40	40-60	Over 60
Expected Frequency	$350(0.15)$ = 52.5	$350(0.73)$ = 255.5	$350(0.06)$ = 21	$350(0.03)$ = 10.5	$350(0.03)$ = 10.5

Do: L1 = 22, 59, 110, 114, 45 and L2 = 52.5, 255.5, 21, 10.5, 10.5

χ^2 GOF-Test(Observed: L1, Expected: L2, df: 4)

$$\chi^2 = \frac{(22-52.5)^2}{52.5} + \frac{(59-255.5)^2}{255.5} + \dots = 1679.6 \quad p\text{-value} \approx 0 \quad df = 4$$

Conclude: Because the p-value (≈ 0) is less than our alpha level (0.05), we reject H_0 . We have convincing evidence that the age distribution for all current users of My Face is different from the distribution a decade ago.

5) A group of volunteers oversees “beautification” for a town, where they maintain flowerbeds, clean up litter, fix broken benches, etc. The town provides the group money, and the items purchased must be allocated as follows:

Category	%
Tools (e.g. shovels, hammers, rakes)	30%
Supplies (e.g. soil, nails, garbage bags)	60%
Food for planning meetings	10%

The group is required to maintain records of all their purchases, and town officials are doing an audit of the 630 items from the past year. In a random sample of 60 purchased items, 15 were tools, 34 were supplies, and 11 were food for planning meetings. Do these observations provide convincing statistical evidence that the volunteers were misallocating money?

State: H_0 : The category distribution for all items purchased this year is the same as the mandated allocation.

H_A : The category distribution for all items purchased this year is not the same as the mandated allocation.

$\alpha = 0.05$

Plan: We will conduct a Chi-square test for goodness of fit, if conditions are met.

Random: Items were randomly selected from all purchases. ✓

10%: $60 \leq 0.10(630 \text{ total items purchased}) \rightarrow 60 \leq 63$ ✓

Large Counts: All expected counts are at least 5. ✓

Age Range	Under 18	18-25	26-40
Expected Frequency	$60(0.3)$ = 18	$60(0.6)$ = 36	$60(0.1)$ = 6

Do: L1 = 15, 34, 11 and L2 = 18, 36, 6

χ^2 GOF-Test(Observed: L1, Expected: L2, df: 2)

$$\chi^2 = \frac{(15-18)^2}{18} + \frac{(34-36)^2}{36} + \dots = 4.778 \quad p\text{-value} = 0.092 \quad df = 2$$

Conclude: Because the p-value (0.092) is greater than our alpha level (0.05), we fail to reject H_0 . We do not have convincing evidence that the distribution of all purchased items differs from the mandated allocation (i.e. we are not convinced that the volunteers were misallocating money).

Further Practice

Teachers: If you'd like to give students additional practice problems...

- Check out our [CED and Textbook Alignment Guide](https://www.skewthescript.org/ap-stats-alignment) ([skewthescript.org/ap-stats-alignment](https://www.skewthescript.org/ap-stats-alignment)) to find additional exercises in your AP Stats textbook or in AP Classroom that are aligned to the content covered in this lesson.
- The following Free Response Questions (FRQs) from [past AP Exams](#) are also aligned to this lesson: 2008 Q5, 2003 Form B Q5 (part c)

